28 Here are the first five terms of a Fibonacci sequence.

3 3 6 9 15

(a) Write down the next two terms of the sequence.

9 + 15 = 24; 15 + 24 = 39

24 , 39

1 mark
Add the previous two terms.

(1)

The first three terms of a different Fibonacci sequence are

a a $2a$

(b) Find the 6th term of this sequence.

4th term: a + 2a = 3a

5th term: 2a + 3a = 5a

6th term: 3a + 5a = 8a

1 mark
Find the 4th and 5th terms.

1 mark
6th term is 8a.

8a

(2)

(Total for Question 28 is 3 marks)

29 $\mathbf{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

Work out $\mathbf{a} - 2\mathbf{b}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$

(Total for Question 29 is 2 marks)

TOTAL FOR PAPER IS 80 MARKS

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